

3. PREDICTIVE MODELLING AND MACHINE LEARNING

Techniques and Best Practices

PRESENTED BY :

VISHNUPRIYA SHAJI



What is Machine Learning?



Machine Learning (ML) is a branch of Artificial Intelligence that enables computers to learn from data and improve their performance without being explicitly programmed for every task.

Machine Learning is the study of algorithms and statistical models that allow computer systems to learn patterns from data and make predictions or decisions automatically.

Types of Machine Learning

1 Supervised Learning

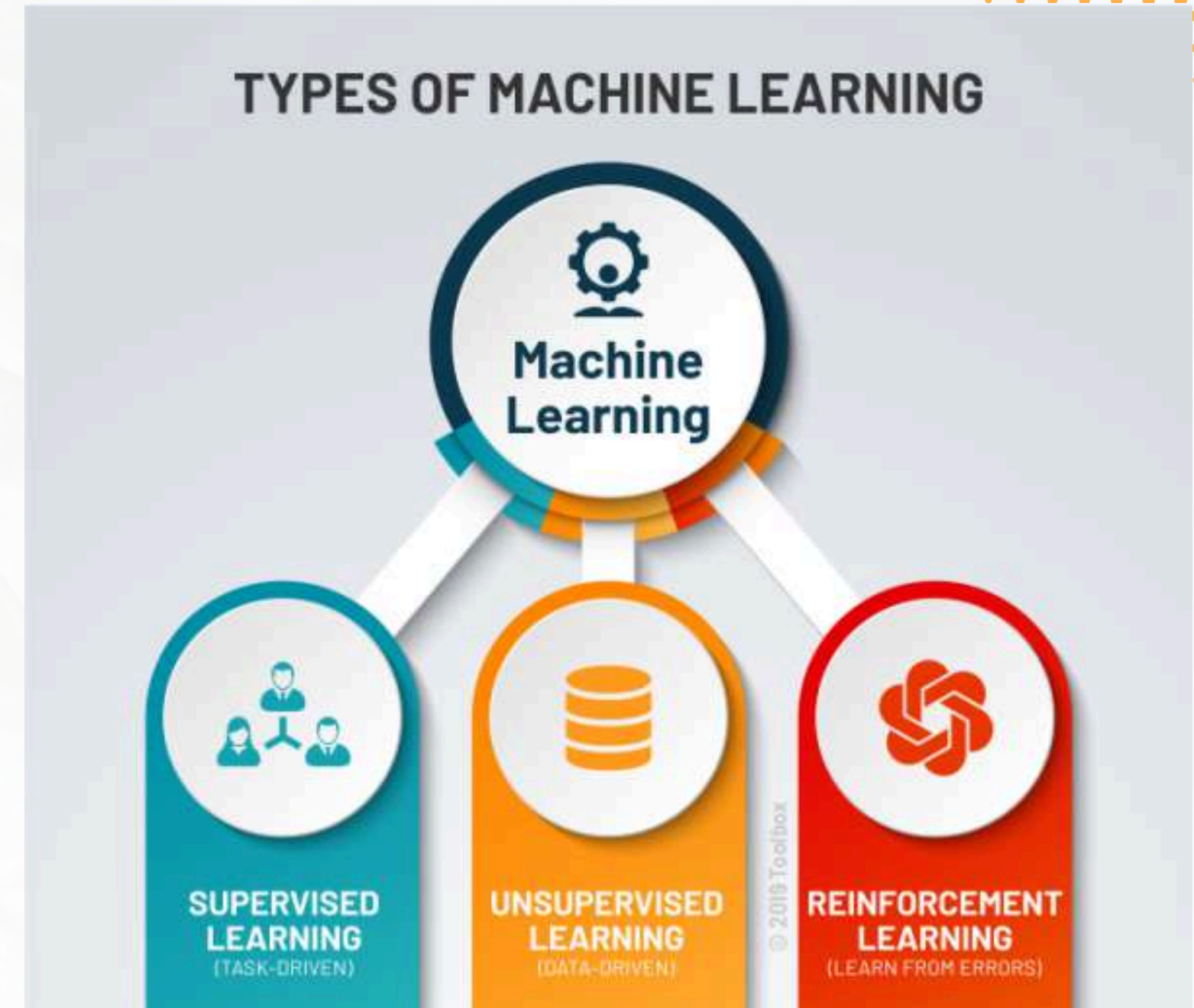
Supervised Learning is a machine learning approach in which a model is trained using labeled data, where each input is associated with a known output, to learn a mapping and make predictions on new data.

2 Unsupervised Learning

Unsupervised Learning is a machine learning approach in which a model is trained using unlabeled data to identify hidden patterns, structures, or relationships within the data.

3 Reinforcement Learning

Reinforcement Learning is a machine learning approach in which an agent learns to make decisions by interacting with an environment and receiving rewards or penalties based on its actions.



Supervised Learning

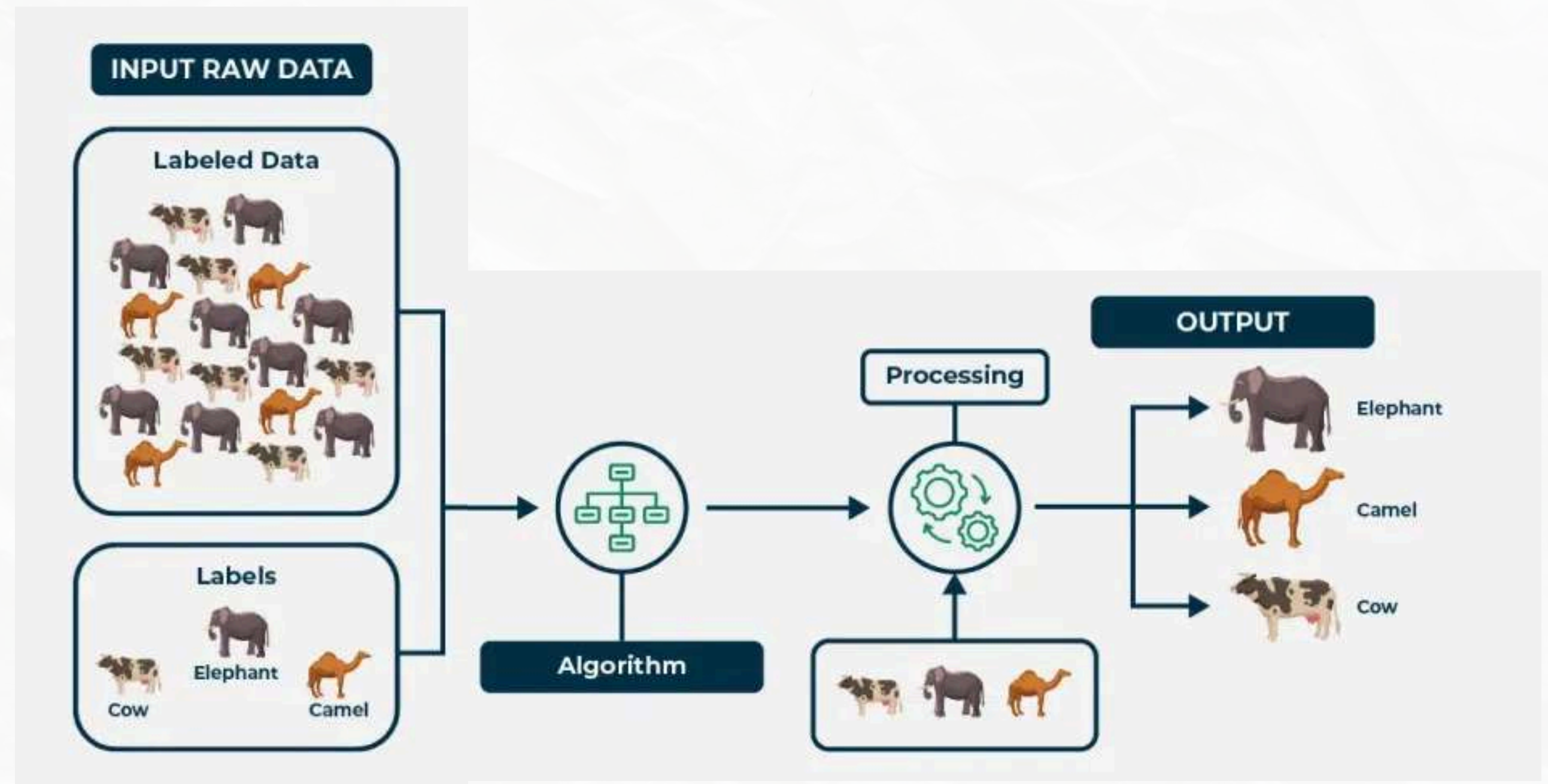
Supervised Learning is a type of machine learning in which the model is trained using labeled data, meaning that each training example contains both input data and the correct output. The algorithm learns the relationship between inputs and outputs and uses this knowledge to predict outcomes for new, unseen data.

Common Algorithms

- Linear Regression
- Logistic Regression
- Decision Tree
- Random Forest
- Support Vector Machine (SVM)
- K-Nearest Neighbors (KNN)
- Naive Bayes

Applications

- House price prediction
- Email spam detection
- Disease diagnosis
- Student performance prediction



Unsupervised Learning

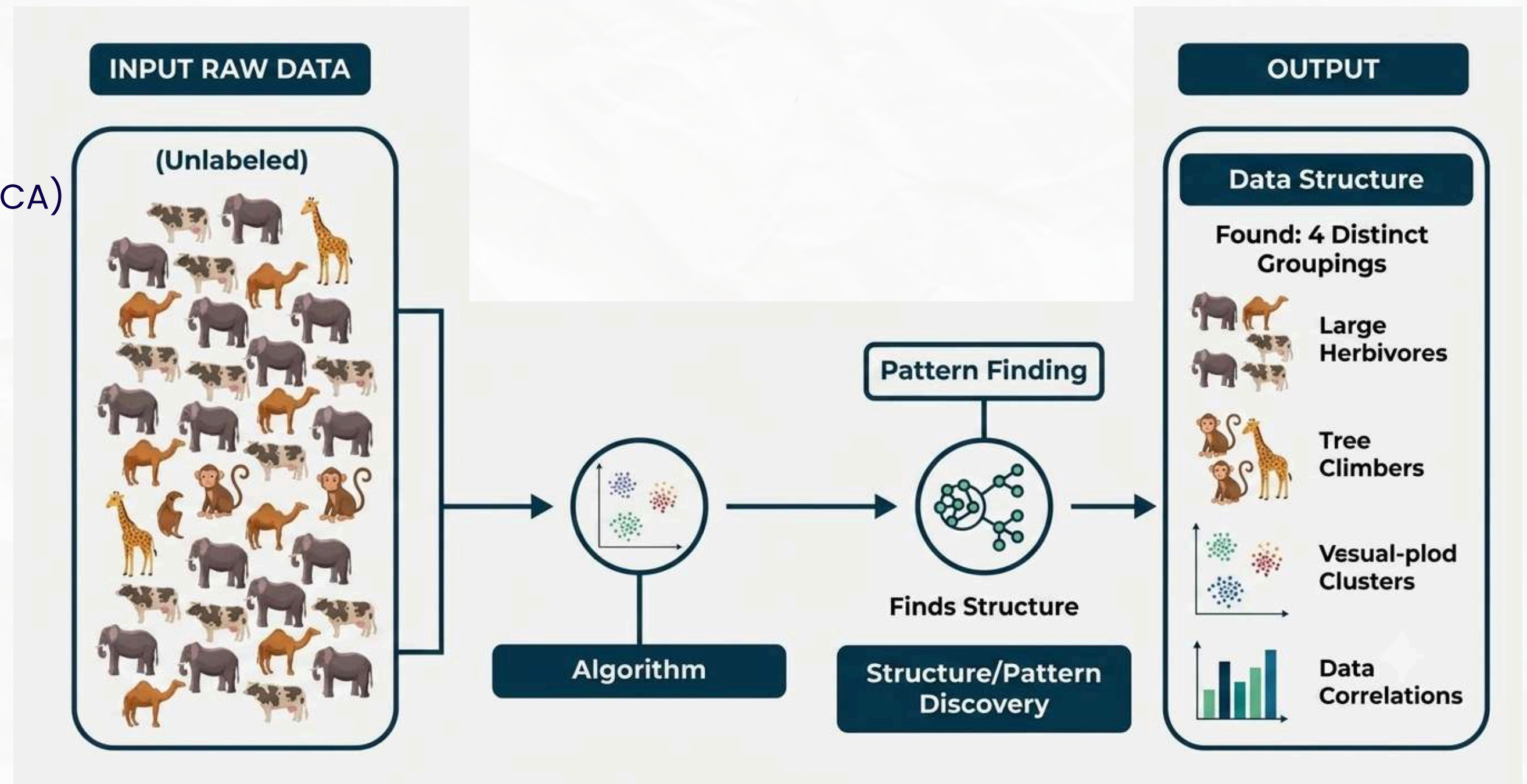
Unsupervised Learning is a type of machine learning in which the model is trained using unlabeled data. The algorithm does not know the correct output and tries to discover hidden patterns, structures, relationships, or groups within the data on its own.

Common Algorithms

- K-Means Clustering
- Hierarchical Clustering
- DBSCAN
- Principal Component Analysis (PCA)
- Apriori Algorithm
- Association Rule Mining

Applications

- Market basket analysis
- Recommendation systems
- Image segmentation
- Social network analysis



Reinforcement Learning

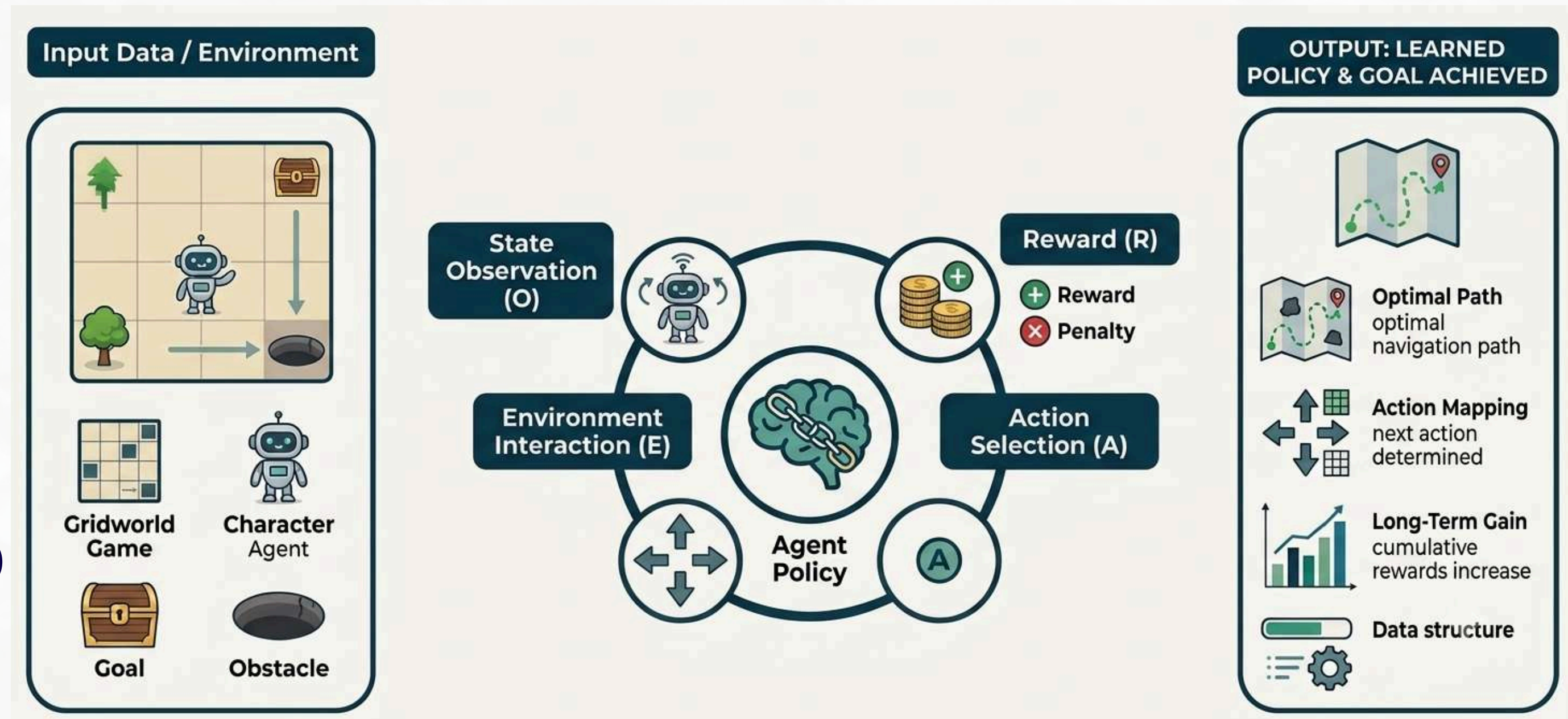
Reinforcement Learning is a type of machine learning in which an agent learns by interacting with an environment and receiving rewards or penalties based on its actions. The goal is to learn the best sequence of actions that maximizes the total reward over time.

Common Algorithms

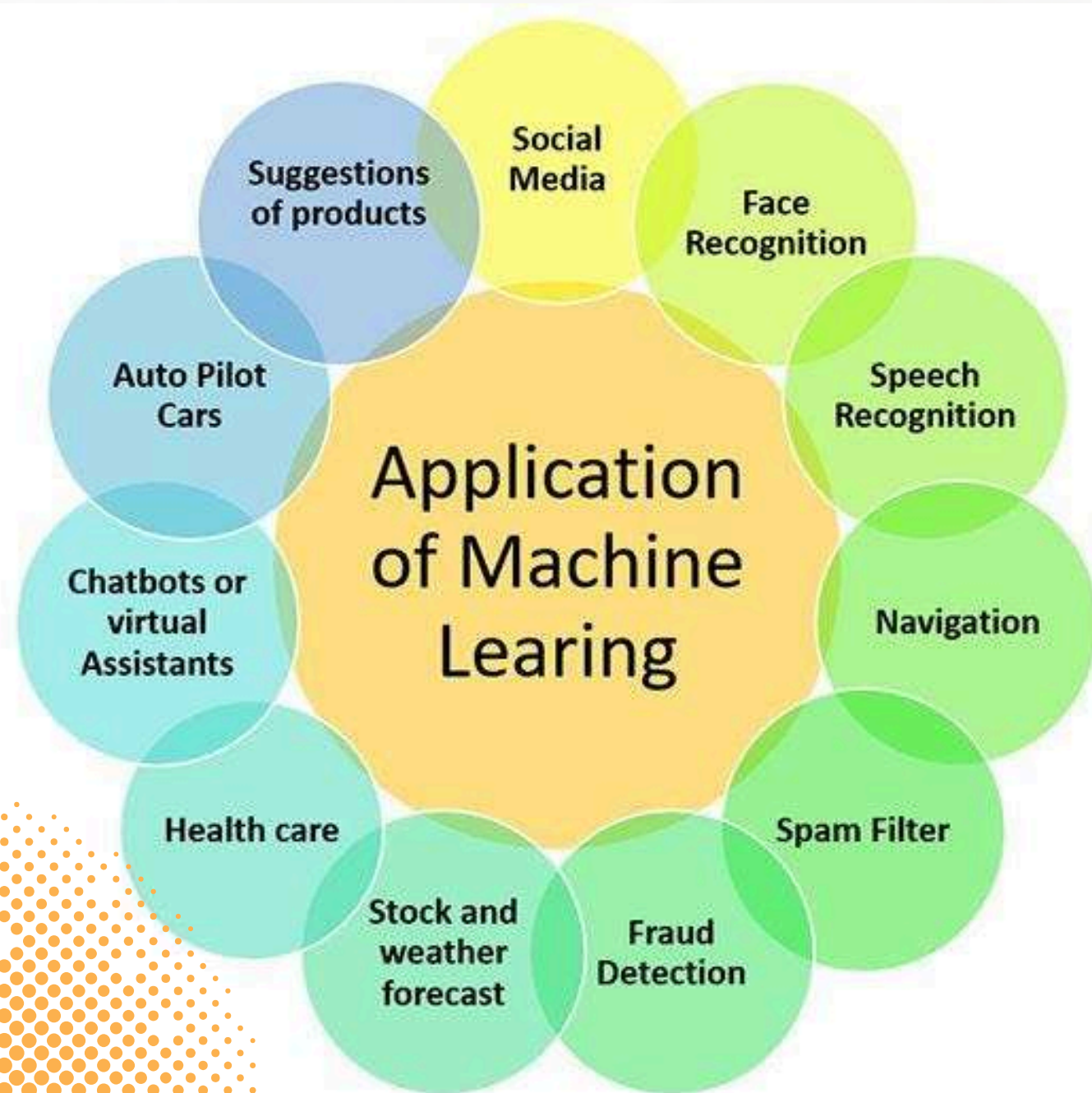
- Q-Learning
- Deep Q Network (DQN)
- SARSA
- Monte Carlo Methods
- Policy Gradient Methods
- Actor-Critic Methods

Applications

- Self-driving cars
- Robotics
- Game playing (Chess, Go)
- Resource allocation



Applications of Machine Learning



1. Healthcare

Machine learning helps in disease diagnosis, medical image analysis, drug discovery, and patient monitoring.

Example: Predicting diabetes based on patient health records.

2. Finance and Banking

Machine learning is used for fraud detection, credit scoring, risk assessment, and stock market analysis.

Example: Detecting suspicious credit card transactions.

3. Education

Machine learning helps analyze student performance, personalize learning, and predict academic outcomes.

Example: Identifying students who are at risk of failing a course.

4. Machine learning enables content recommendation, sentiment analysis, and targeted advertising.

Example: Recommending posts and videos based on user interests.

Introduction to Statistical Modelling

Statistical Modelling is the process of using mathematical and statistical techniques to represent, analyze, and understand relationships between variables in data.

It helps in:

- Understanding patterns in data
- Making predictions
- Testing hypotheses
- Supporting decision-making

Example

A company wants to predict sales based on advertising expenditure. A statistical model can identify the relationship between advertising cost and sales and predict future sales.



Model Building Process



Model Building Process

1. Problem Definition

Problem Definition is the process of clearly identifying the objective of the model and the problem it is intended to solve. It defines the expected outcome and guides the entire model-building process.

Example

A college wants to identify students who are likely to fail so that extra support can be provided before examinations.

Output

Goal: Predict whether a student will Pass or Fail.

2. Data Collection

Data Collection is the process of gathering relevant data from various sources that can help solve the defined problem.

Example

The college collects:

- Attendance
- Internal Marks ,etc

Output

A dataset containing information about students.

3. Data Preprocessing

Data Preprocessing is the process of cleaning, transforming, and organizing raw data to make it suitable for analysis and model training.

Example

The collected student data may contain:

- Missing attendance records
- Incorrect marks

These issues are corrected before modeling.

Output

Clean and reliable student data

4. Exploratory Data Analysis (EDA)

Exploratory Data Analysis is the process of examining and visualizing data to understand its characteristics, patterns, trends, and relationships.

Example

The college analyzes:

- Do students with higher attendance score better?
- Does study time affect performance?

A scatter plot may show that students with higher attendance usually pass.

5. Feature Selection

Feature Selection is the process of choosing the most relevant variables for model building, while Feature Engineering involves creating new meaningful features from existing data.

Example

Selected Features:

- Attendance
- Internal Marks
- Study Hours

Created Feature:

- Total Performance Score = Internal Marks + Assignment Marks

Output

Important variables for prediction.

6. Model Selection

Model Selection is the process of choosing the most appropriate algorithm to solve the problem based on the nature of the data and objective. Example

Example

Since the goal is Pass/Fail prediction (Classification), the college chooses:

- Decision Tree
- Logistic Regression

Output

Selected machine learning algorithm.

7. Data Splitting

Data Splitting is the process of dividing the dataset into training and testing datasets to evaluate model performance on unseen data.

Example

1000 Student Records

- 800 → Training Data
- 200 → Testing Data

Output

Training and Testing datasets.

8. Model Training

Model Training is the process in which the selected algorithm learns patterns and relationships from the training data.

Example

The model learns that:

- High attendance + High marks → Pass
- Low attendance + Low marks → Fail

Output

A trained prediction model.

9. Model Evaluation

Model Evaluation is the process of measuring the performance of the trained model using testing data and appropriate evaluation metrics.

Example

The model predicts outcomes for 200 unseen students.

Actual Result vs Predicted Result is compared.

If 180 predictions are correct:

Accuracy = 90%

Output

Performance metrics such as Accuracy, Precision, Recall, and F1-Score.

10. Model Deployment

Model Deployment is the process of integrating the trained model into a real-world environment where it can be used to make predictions.

Example

The college integrates the model into its student management system.

When a student's attendance and marks are entered, the system predicts:

- Pass
- Fail
- At Risk

Output

A working prediction system.

11. Monitoring and Maintenance

Monitoring and Maintenance is the continuous process of tracking model performance and updating the model when necessary to maintain accuracy and reliability.

Example

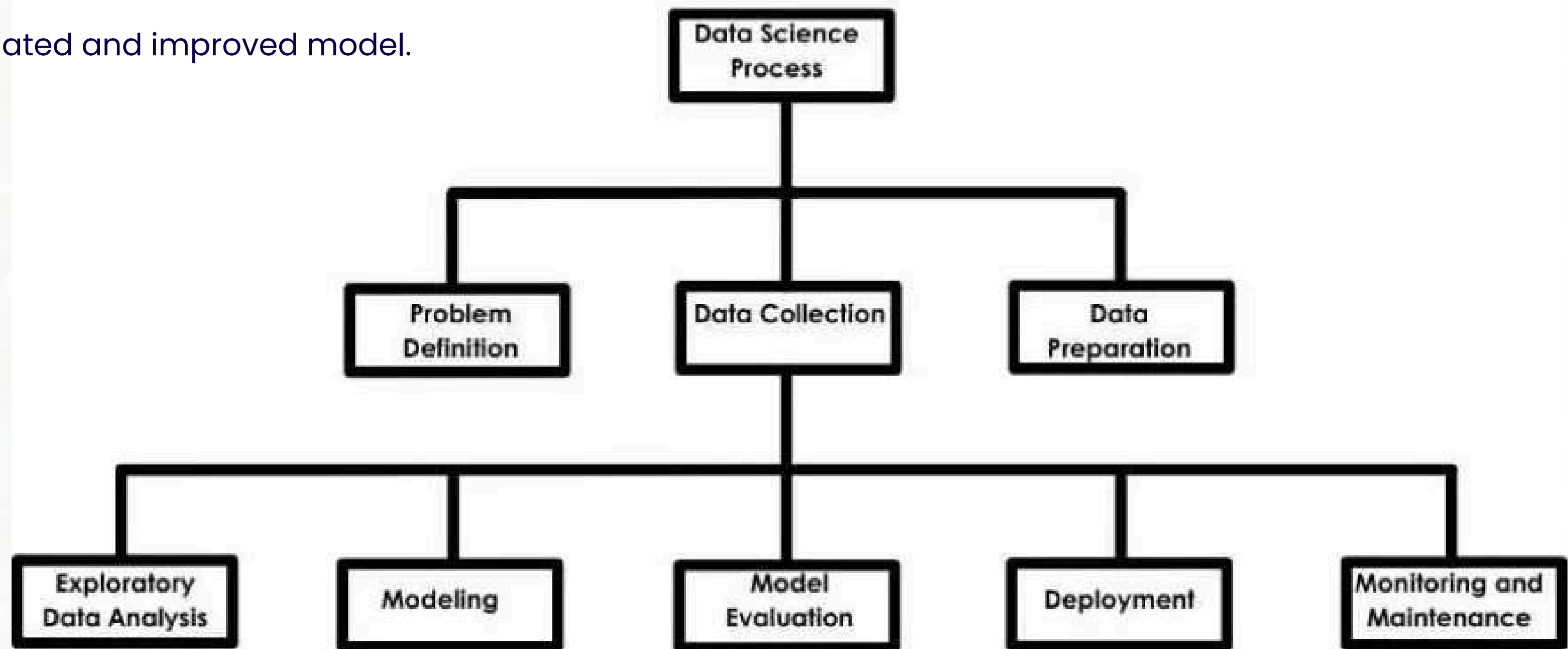
After one year, the examination pattern changes.

The model's accuracy drops from 90% to 75%.

The college collects new student data and retrain the model.

Output

An updated and improved model.



Training and Testing Data Concepts

When building a machine learning model, it is important to evaluate how well the model performs on new, unseen data. To achieve this, the dataset is divided into two parts:

1. **Training Data**

2. **Testing Data**

This process helps determine whether the model has learned meaningful patterns or simply memorized the data.

Training Data

Training data is the dataset used to teach a machine learning model. It usually contains labeled examples (where the correct output is already known). The model studies these examples, finds patterns, and slowly learns to make predictions on its own.

During training, the model:

- looks at input and output pairs
- identifies relationships
- adjusts its internal rules
- improves its accuracy over time

Models with large and good-quality training data usually perform better.

Testing Data

Once the model has learned from training data, we need new, unseen data to check if it has learned correctly. This new dataset is called testing data. Testing data helps to:

- measure accuracy
- check if the model is overfitting
- verify if the model can handle new information

If a model performs well on testing data, it means it has truly understood the patterns instead of just memorizing.

A common split is **80%** training and **20%** testing.

Split Ratio	Training Data	Testing Data
80:20	80%	20%
70:30	70%	30%
75:25	75%	25%

Simple Linear Regression



Simple Linear Regression is a statistical method used to model the linear relationship between a dependent variable and a single independent variable by fitting a straight line to the observed data.

Example

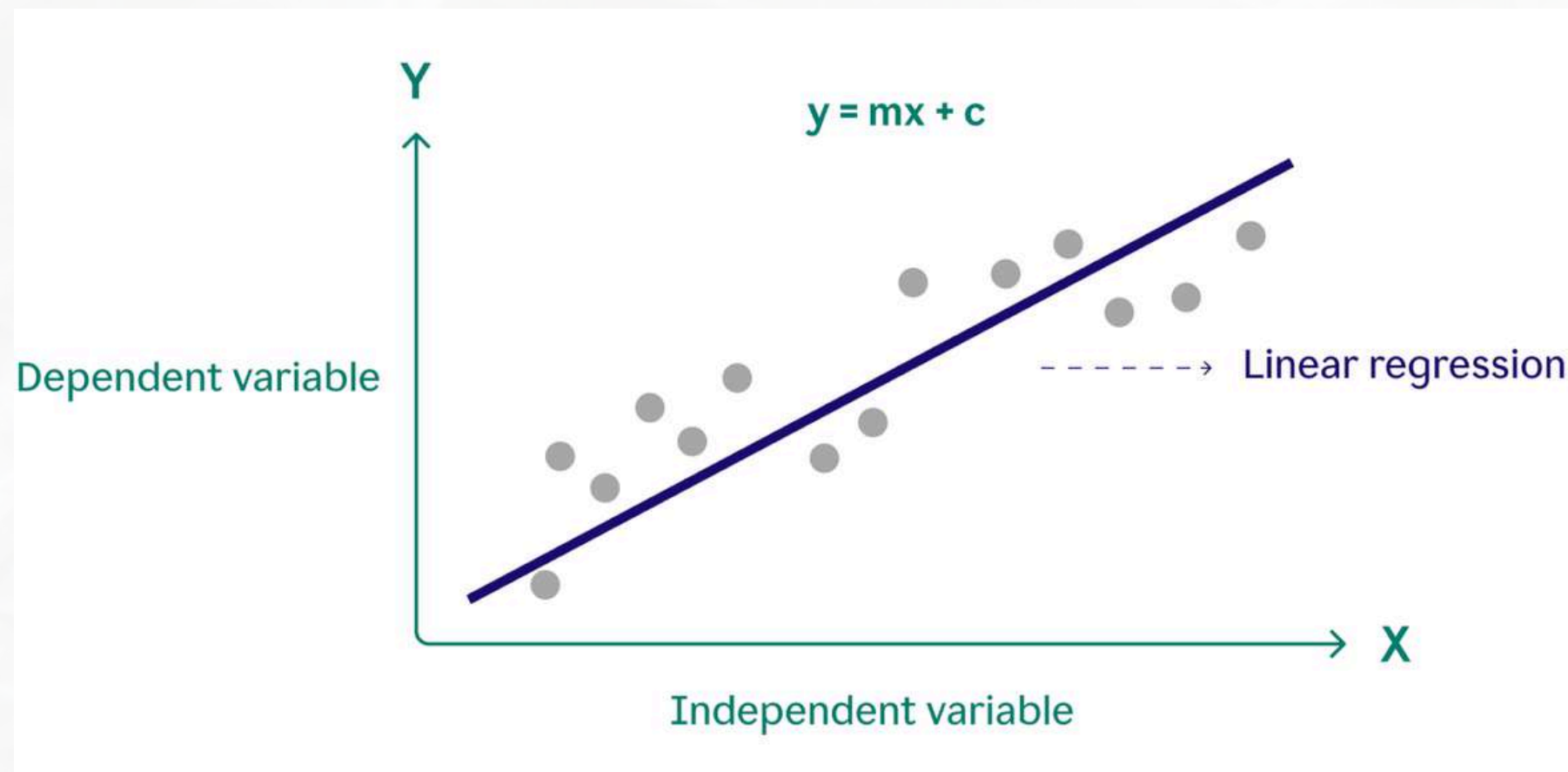
A college wants to predict a student's exam score based on the number of hours studied.

- Independent Variable (X) = Study Hours
- Dependent Variable (Y) = Exam Score

As study hours increase, exam scores tend to increase.

It helps us understand how changes in one variable affect another and enables us to make predictions.

Model Formulation



The relationship between the dependent and independent variables is represented by the simple linear equation:

Simple Linear Regression Formula:- **$y = mx + c$**

Here:

- y is the predicted value (dependent variable).
- m is the slope of the line
- x is the independent variable.
- c is the y -intercept (the value of y when x is 0).

In this equation m signifies the slope of the line indicating how much y changes for a one-unit increase in x , a positive m suggests a direct relationship while a negative m indicates an inverse relationship.

Solved Example 1

A teacher wants to predict a student's Exam Score (Y) based on Study Hours (X).

Step 1: Find the Mean of X and Y

$$\bar{X} = \frac{2 + 4 + 6 + 8 + 10}{6}$$

$$\bar{X} = 6$$

$$\bar{Y} = \frac{30 + 40 + 50 + 60 + 70}{6}$$

$$\bar{Y} = 50$$

Step 2: Create the Calculation Table

X	Y	X - $\bar{X}$	Y - $\bar{Y}$	(X - $\bar{X}$)(Y - $\bar{Y}$)	(X - $\bar{X}$) ²
2	30	-4	-20	80	16
4	40	-2	-10	20	4
6	50	0	0	0	0
8	60	2	10	20	4
10	70	4	20	80	16
Total		0	0	200	40

Student Study Hours	Exam (X)	Score (Y)
A	2	30
B	4	40
C	6	50
D	8	60
E	10	70

Step 3: Calculate the Slope (m)=

$$m = \frac{\Sigma (X - \bar{X}) - (Y - \bar{Y})}{\Sigma (X - \bar{X})^2} \quad m = \frac{200}{40} = 5$$

Slope (m) = 5

Step 4: Calculate the Intercept (c)=

$$c = \bar{Y} - m\bar{X}$$
$$c = 50 - (5 \times 6)$$

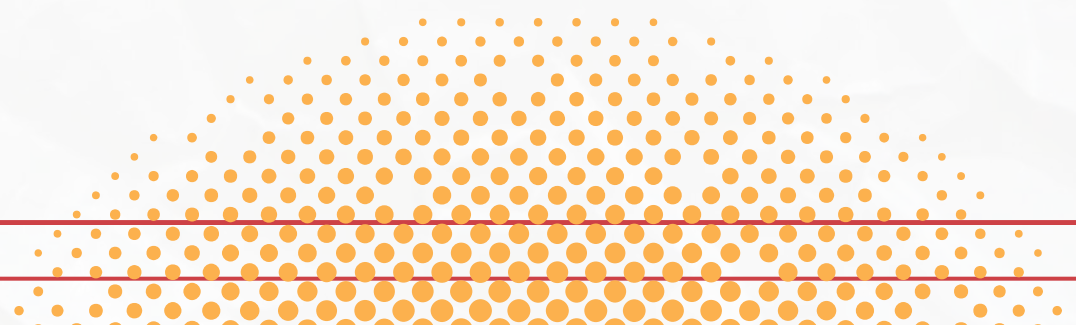
Intercept (c) = 20

Step 5: Form the Regression Equation

$$Y = mx + c$$
$$Y = 5(X) + 20$$

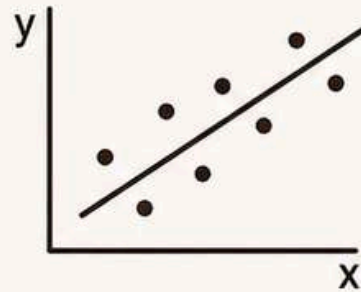
Step 6: Make a Prediction

Suppose a student studies 7 hours.
Substitute X=7:

$$Y = (5 \times 7) + 20 \quad Y = 55$$
$$55 = 35 + 20$$


Assumptions Of Linear Regression

1. LINEARITY



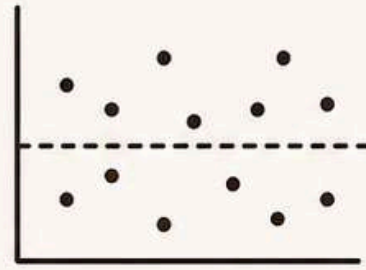
The relationship between X and Y is linear.

2. INDEPENDENCE OF ERRORS



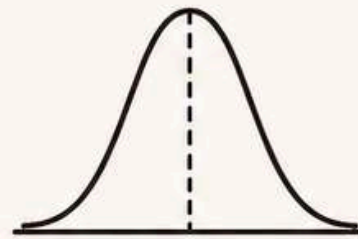
The errors in predictions should not affect each other.

3. HOMOSCEDASTICITY (CONSTANT VARIANCE)



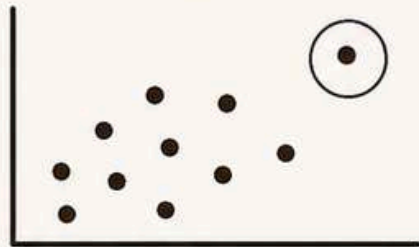
The errors should have equal spread across all values of X.

4. NORMALITY OF ERRORS



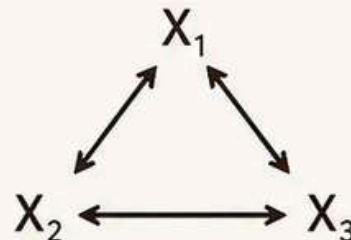
The errors should follow a normal (bell-shaped) distribution.

5. NO SIGNIFICANT OUTLIERS



The data should not contain extreme values that strongly influence the regression line.

6. NO MULTICOLLINEARITY (MULTIPLE REGRESSION)



Independent variables should not be highly correlated with each other.

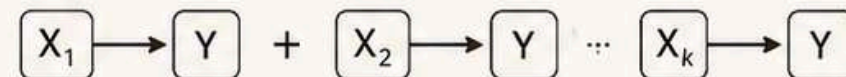
7. NO AUTOCORRELATION (TIME SERIES DATA)



Errors should not be correlated with each other over time; there should be no pattern.

8. ADDITIVITY (MULTIPLE REGRESSION)

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_k X_k + \varepsilon$$



The total effect on Y is the sum of individual effects from each X, with no interaction unless explicitly included.

1. Linearity: The relationship between inputs (X) and the output (Y) is a straight line.

2. Independence of Errors: The errors in predictions should not affect each other.

3. Constant Variance: The errors should have equal spread across all values of the input. If the spread changes (like fans out or shrinks), it's called heteroscedasticity and it's a problem for the model.

4. Normality of Errors: The errors should follow a normal (bell-shaped) distribution

5. No Multicollinearity (for multiple regression): Input variables shouldn't be too closely related to each other.

6. No Autocorrelation: Errors shouldn't show repeating patterns, especially in time-based data

7. Additivity: The total effect on Y is just the sum of effects from each X, no mixing or interaction between them

Interpretation of Regression Coefficients

In a Simple Linear Regression model, the regression coefficients describe the relationship between the independent variable (X) and the dependent variable (Y). They indicate how the dependent variable changes when the independent variable changes.

1. Intercept (c)

The intercept (c) is the predicted value of the dependent variable (Y) when the independent variable (X) is equal to zero. It represents the starting point of the regression line.

2. Slope (m)

The slope (m), also called the regression coefficient, represents the average change in the dependent variable (Y) for every one-unit increase in the independent variable (X).

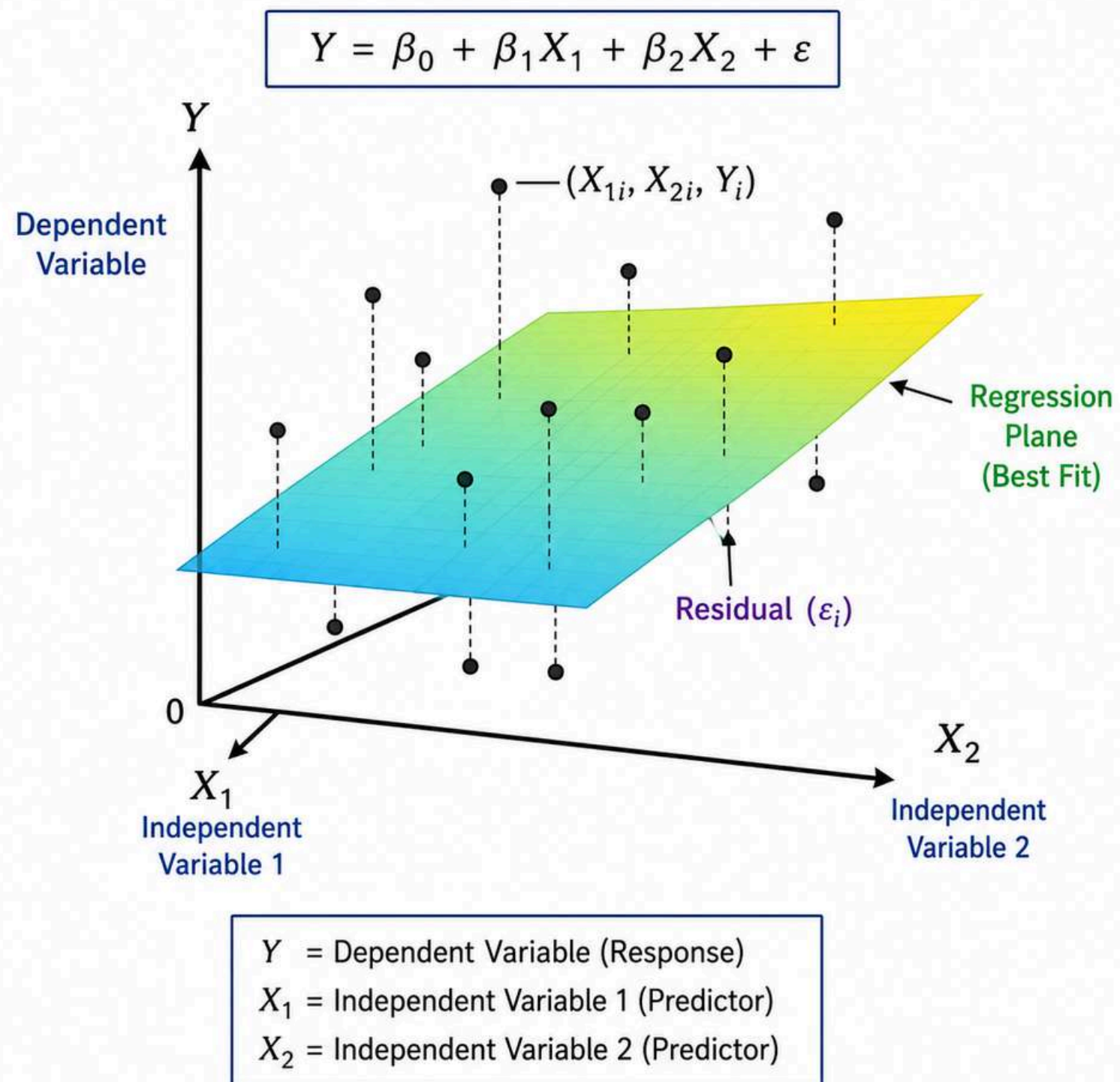
3. Positive Regression Coefficient

A positive regression coefficient indicates that the dependent variable increases as the independent variable increases.

4. Negative Regression Coefficient

A negative regression coefficient indicates that the dependent variable decreases as the independent variable increases.

Multiple Linear Regression



Multiple Linear Regression is a statistical technique that examines the relationship between one dependent variable and two or more independent variables.

It is used to predict the value of a dependent variable by considering the combined effect of multiple predictor variables.

Example

A college wants to predict a student's Exam Score based on:

- Study Hours
- Attendance Percentage
- Assignment Marks

Instead of using only one factor (study hours), the model considers all three factors to make a more accurate prediction.

Model Formulation

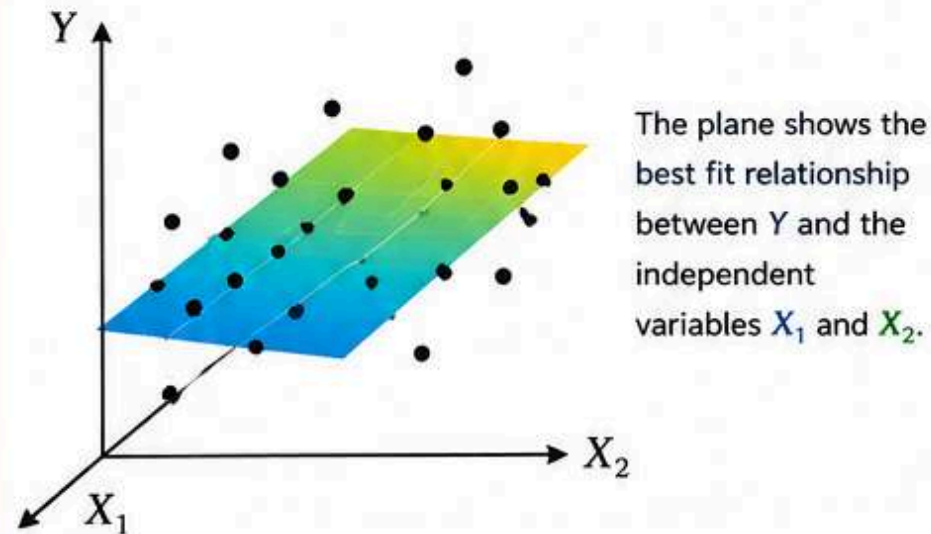
MODEL FORMULATION

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_k X_k + \varepsilon$$

Where:

- Y = Dependent variable (response)
- $X_1, X_2, \dots, X_k$ = Independent variables (predictors)
- β_0 = Intercept (constant term)
- $\beta_1, \beta_2, \dots, \beta_k$ = Regression coefficients (slopes)
- ε = Error term (residual)

GEOMETRIC REPRESENTATION ($k = 2$)



INTERPRETATION OF COEFFICIENTS

The coefficient β_j represents the expected change in Y for a one-unit increase in X_j , holding all other independent variables constant.

Example ($k = 2$):

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2$$

- β_1 : Change in Y for a one-unit increase in X_1 , keeping X_2 constant.
- β_2 : Change in Y for a one-unit increase in X_2 , keeping X_1 constant.

EXAMPLE

Suppose we want to predict House Price (Y) using Size in sq.ft (X_1) and Number of Bedrooms (X_2).

Model:

$$\text{Price} = \beta_0 + \beta_1(\text{Size}) + \beta_2(\text{Bedrooms}) + \varepsilon$$

- β_1 : Expected change in Price for 1 sq.ft increase in Size, keeping Bedrooms constant.
- β_2 : Expected change in Price for 1 more Bedroom, keeping Size constant.

The mathematical model for Multiple Linear Regression is:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \varepsilon$$

Where,

- $Y \rightarrow$ Variable to be predicted.
- $X_1, X_2, X_3 \dots \rightarrow$ Variables used for prediction.
- $\beta_0 \rightarrow$ Predicted value when all independent variables are zero.
- $\beta_1, \beta_2, \beta_3 \dots \rightarrow$ Change in Y for a one-unit increase in the corresponding X while keeping all other variables constant.
- $\varepsilon \rightarrow$ Random error not explained by the model.

Example

A college wants to predict a student's Final Exam Score based on:

- Study Hours (X_1)
- Attendance (%) (X_2)
- Assignment Marks (X_3)

The regression equation is:

$$\text{Score} = 10 + 4(\text{Study Hours}) + 0.3(\text{Attendance}) + 0.5(\text{Assignment Marks})$$

Interpretation of Regression Coefficients

In Multiple Linear Regression (MLR), each regression coefficient represents the effect of one independent variable on the dependent variable while keeping all other independent variables constant.

1. Interpretation of the Intercept (β_0)

The intercept (β_0) is the predicted value of the dependent variable when all independent variables are equal to zero. It represents the baseline value of the regression model.

2. Interpretation of β_1 (Study Hours)

β_1 represents the expected change in the dependent variable for every one-unit increase in Study Hours, while keeping Attendance and Assignment Marks constant.

3. Interpretation of β_2 (Attendance)

β_2 represents the expected change in the dependent variable for every one-unit increase in Attendance, while keeping the other variables constant.

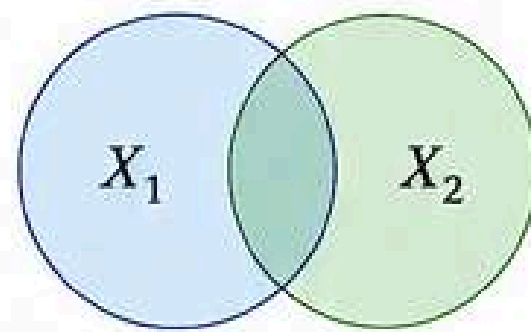
4. Interpretation of β_3 (Assignment Marks)

β_3 represents the expected change in the dependent variable for every one-unit increase in Assignment Marks, while keeping all other variables constant.

Multicollinearity

WHY DOES IT OCCUR?

- When two or more predictors provide similar information.
- When one predictor is a linear combination of others.
- When variables are measured in similar ways.
- When data is collected over a narrow range.



High correlation
between X_1
and X_2

Employee	Years of Experience (X_1)	Age (X_2)	Salary (Y)
A	2	24	30,000
B	4	26	40,000
C	6	28	50,000
D	8	30	60,000
E	10	32	70,000

Multicollinearity is a condition in Multiple Linear Regression in which two or more independent variables are highly correlated, this means that one independent variable can be predicted from another because they contain similar information, making it difficult to estimate the individual effect of each variable to the dependent variable.

Example

Suppose a company wants to predict an employee's Salary using the following independent variables:

- Years of Experience
- Age

Here, **Age** and **Years of Experience** increase together. As employees become older, they generally gain more work experience.

Therefore, the two independent variables are highly correlated. The model cannot easily determine whether the increase in salary is mainly due to Age or Years of Experience.

This situation is called **Multicollinearity**.

Causes and Effects of Multicollinearity

EFFECTS OF MULTICOLLINEARITY



Regression coefficients become unstable and unreliable.



Standard errors of coefficients increase.



It becomes difficult to determine the individual significance of each predictor.



Small changes in data can cause large changes in coefficients.



Model interpretation becomes misleading.

Causes of Multicollinearity

1. Including highly correlated independent variables.
2. Creating new variables that are combinations of existing variables.
3. Collecting data with similar or duplicate features.
4. Using too many predictor variables in the model.

Effects of Multicollinearity

1. Makes regression coefficients unstable.
2. Increases the standard errors of the coefficients.
3. Makes it difficult to interpret the effect of each independent variable.
4. Small changes in the data can produce large changes in coefficient values.
5. Reduces the reliability of the regression model.

Detecting Multicollinearity

1. CORRELATION MATRIX

Check pairwise correlation between independent variables.
Correlation close to +1 or -1 indicates multicollinearity.

Variable	Experience (X_1)	Age (X_2)
Experience (X_1)	1.00	0.95
Age (X_2)	0.95	1.00

Since 0.95 is close to +1, multicollinearity exists.

2. VARIANCE INFLATION FACTOR (VIF)

Measures how much the variance of a coefficient is inflated.

$$VIF = \frac{1}{1 - R^2}$$

where R^2 = Coefficient of determination regressing one variable on the others.

Interpretation:

VIF Value	Interpretation
1	No multicollinearity
1 – 5	Low to moderate
> 5	High (possible concern)
> 10	Severe multicollinearity

3. TOLERANCE

Another measure used.

$$\text{Tolerance} = 1 - R^2$$

Interpretation:

Tolerance Value	Interpretation
Close to 1	No multicollinearity
0.2 – 1	Moderate
< 0.2 (or < 0.1)	High multicollinearity

1. Correlation Matrix

A correlation matrix shows the correlation between independent variables.

- Correlation close to +1 or -1 indicates strong correlation.
- Strong correlation suggests multicollinearity.

2. Variance Inflation Factor (VIF)

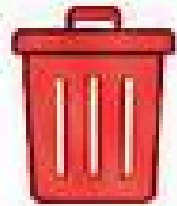
The Variance Inflation Factor (VIF) measures how much the variance of a regression coefficient is inflated due to multicollinearity.

3. Tolerance

Tolerance is another measure used to detect multicollinearity.

Methods to Handle Multicollinearity

HOW TO HANDLE MULTICOLLINEARITY?

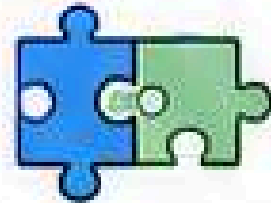


1. REMOVE HIGHLY CORRELATED VARIABLES

Remove one of the highly correlated variables that provides similar information.

Example:

Keep either Age or Experience, not both.

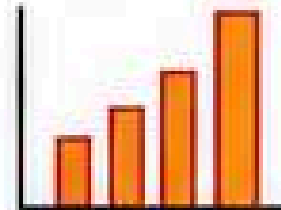


2. COMBINE VARIABLES

Combine correlated variables into a single variable.

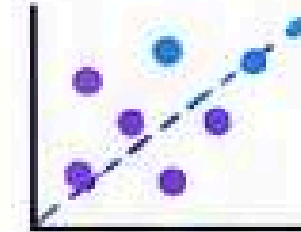
Example:

Combine Midterm Marks and Final Marks into Overall Performance.



3. COLLECT MORE DATA

Increasing the number of observations may reduce the effect of multicollinearity.



4. APPLY PCA

Use Principal Component Analysis to transform correlated variables into uncorrelated components.



5. USE REGULARIZATION TECHNIQUES

Techniques like Ridge, Lasso, or Elastic Net can reduce the impact.

Example:

Ridge Regression, Lasso Regression

If two variables provide similar information, remove one of them.

Example: Keep either Age or Years of Experience, but not both.

Merge correlated variables into a single feature.

Example: Combine Midterm Marks and Final Marks into Overall Academic Performance.

Increasing the number of observations can sometimes reduce the impact of multicollinearity.

PCA transforms correlated variables into a smaller set of new, uncorrelated variables called principal components.

Regularization methods reduce the impact of multicollinearity by shrinking coefficient values.

Key Points for Multicollinearity

A company wants to predict Salary (Y) using Years of Experience (X_1) and Age (X_2).

Employee	Years of Experience (X_1)	Age (X_2)	Salary (Y)
A	2	24	30,000
B	4	26	40,000
C	6	28	50,000
D	8	30	60,000
E	10	32	70,000

Observation:

As experience increases, age also increases.

Experience and Age are highly correlated.

Including both in the model will cause multicollinearity.

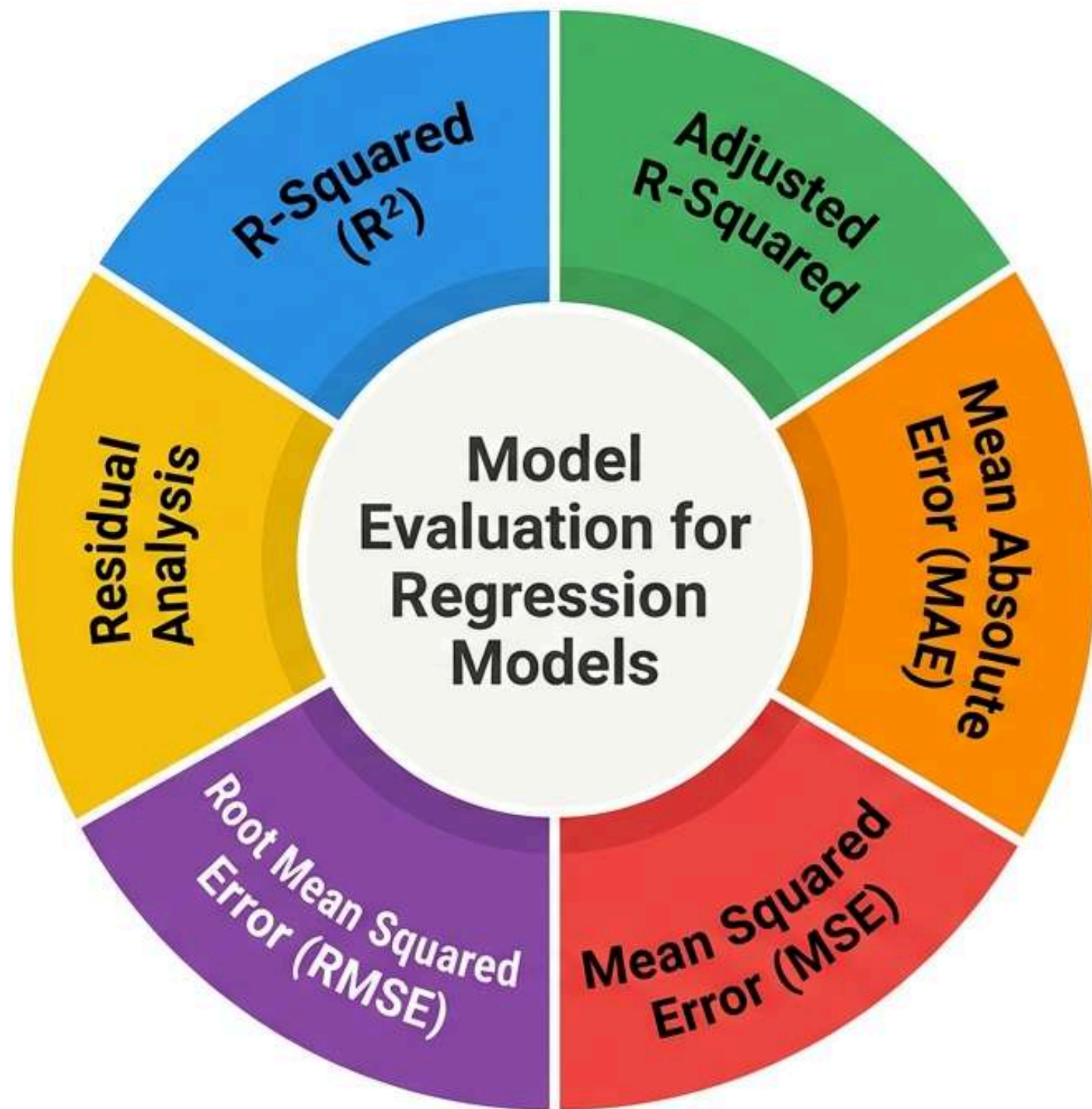
1. Multicollinearity occurs only in Multiple Linear Regression, as it requires two or more independent variables.
2. It does not affect the model's ability to make predictions directly but makes the regression coefficients unreliable and difficult to interpret.
3. The most common methods to detect multicollinearity are the Correlation Matrix, Variance Inflation Factor (VIF), and Tolerance.
4. It can be reduced by removing correlated variables, combining features, collecting more data, using PCA, or applying Ridge or Lasso Regression.



SUMMARY:

Multicollinearity is the presence of high correlation among independent variables, which reduces the reliability and interpretability of the regression model.

Model Evaluation for Regression Models



Model evaluation is the process of measuring how well a regression model predicts the dependent variable. It helps determine the model's accuracy, reliability, and ability to generalize to new data.

The most commonly used evaluation metrics are:

1. R-Squared (R^2)
2. Adjusted R-Squared
3. Mean Absolute Error (MAE)
4. Mean Squared Error (MSE)
5. Root Mean Squared Error (RMSE)
6. Residual Analysis

1. R-SQUARED (R^2)

Measures the proportion of variance in the dependent variable explained by the model.

$$R^2 = 1 - \frac{\sum_{i=1}^n (Y_i - \hat{Y}_i)^2}{\sum_{i=1}^n (Y_i - \bar{Y})^2}$$

Where:

Y_i = Actual value

$\hat{Y}_i$ = Predicted value

$\bar{Y}$ = Mean of actual values

n = Number of observations

Range: $0 \leq R^2 \leq 1$ (Higher is better)

2. ADJUSTED R-SQUARED

Adjusted R^2 adjusts the R^2 value for the number of predictors in the model.

$$\text{Adjusted } R^2 = 1 - \frac{(1 - R^2)(n - 1)}{n - p - 1}$$

Where:

R^2 = R-Squared

n = Number of observations

p = Number of independent variables (predictors)

Range: Can be negative, but higher is better

3. MEAN ABSOLUTE ERROR (MAE)

Average of the absolute differences between actual and predicted values.

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |Y_i - \hat{Y}_i|$$

Where:

Y_i = Actual value

$\hat{Y}_i$ = Predicted value

n = Number of observations

Range: 0 to ∞ (Lower is better)

4. MEAN SQUARED ERROR (MSE)

Average of the squared differences between actual and predicted values.

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (Y_i - \hat{Y}_i)^2$$

Where:

Y_i = Actual value

$\hat{Y}_i$ = Predicted value

n = Number of observations

Range: 0 to ∞ (Lower is better)

5. ROOT MEAN SQUARED ERROR (RMSE)

Square root of MSE. It measures the average prediction error in the same units as Y .

$$\begin{aligned} \text{RMSE} &= \sqrt{\frac{1}{n} \sum_{i=1}^n (Y_i - \hat{Y}_i)^2} \\ &= \sqrt{\text{MSE}} \end{aligned}$$

Where:

Y_i = Actual value

$\hat{Y}_i$ = Predicted value

n = Number of observations

Range: 0 to ∞ (Lower is better)

6. RESIDUAL

Residual is the difference between actual and predicted values.

$$e_i = Y_i - \hat{Y}_i$$

Where:

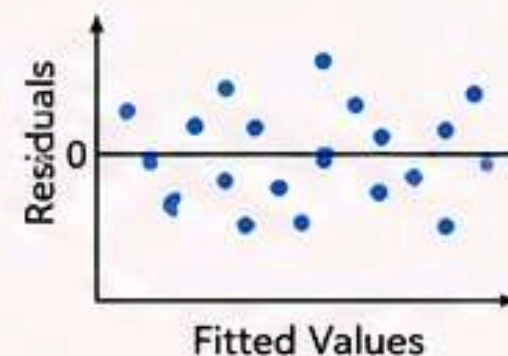
e_i = Residual for i^{th} observation

Y_i = Actual value

$\hat{Y}_i$ = Predicted value

Residual Analysis:

- Residuals should be randomly scattered around zero.
- No pattern should be present.
- Constant spread indicates good model fit.



1. R-Squared (R^2)

R-Squared (Coefficient of Determination) measures the proportion of the variance in the dependent variable that is explained by the independent variables. It indicates how well the regression model fits the data.

In Simple Words,

R^2 tells us how well the regression model explains the data. It shows what percentage of the variation in the actual values is explained by the model.

2. Adjusted R-Squared

Adjusted R^2 adjusts the R^2 value by taking into account the number of predictors and observations in the regression model.

In Simple Words,

Adjusted R^2 is an improved version of R^2 that considers how many independent variables are used in the model. It prevents unnecessary variables from making the model appear better than it really is.

3. Mean Absolute Error (MAE)

Mean Absolute Error is the average of the absolute differences between actual and predicted values.

In Simple Words,

MAE tells us the average prediction error made by the model, ignoring whether the error is positive or negative.

4. Mean Squared Error (MSE)

Mean Squared Error is the average of the squared differences between actual and predicted values.

5. Root Mean Squared Error (RMSE)

Root Mean Squared Error is the square root of the Mean Squared Error.

6. Residual Analysis

Residual Analysis is the process of examining the residuals (actual value – predicted value) to verify whether the assumptions of linear regression are satisfied.

In Simple Words,

Residual analysis checks whether the prediction errors are random. If they are randomly scattered around zero, the regression model is considered appropriate.

Solved Example 2

Step 1:- Calculate the Mean of Actual Marks

Step 2: Create the Calculation Table

*Note:- $(Y - \bar{Y})$ = Actual Marks - Mean of Actual Marks

Student	A	B	C	D	E
Actual Marks (Y)	50	60	70	80	90
Predicted Marks ($\hat{Y}$)	48	63	68	82	89

Y_i = Actual value of the dependent variable

$\hat{Y}_i$ = Predicted value from the regression model

$\bar{Y}$ = Mean of actual values

n = Number of observations

p = Number of independent variables

e_i = Residual (error term) for i^{th} observation

Student	Actual Marks (Y)	Predicted Marks ($\hat{Y}$)	Residual (Y- $\hat{Y}$)	Residual ²	(Y- $\bar{Y}$)	(Y- $\bar{Y}$) ²
A	50	48	2	4	-20	400
B	60	63	-3	9	-10	100
C	70	68	2	4	0	0
D	80	82	-2	4	10	100
E	90	89	1	1	20	400
Total				22		1000

1. R-Squared (R^2)

$$R^2 = 1 - \frac{\sum (Y_i - \hat{Y}_i)^2}{\sum (Y_i - \bar{Y})^2}$$

$$R^2 = 1 - \frac{22}{1000}$$

Therefore,

$$R^2 = 0.978$$

Interpretation

The model explains 97.8% of the variation in marks.

Only 2.2% of the variation is unexplained.

2. Adjusted R-Squared

$$\text{Adjusted } R^2 = 1 - \frac{(1 - R^2)(n - 1)}{n - p - 1}$$

$$\text{Adjusted } R^2 = 1 - \frac{(1 - 0.978)(5 - 1)}{5 - 2 - 1}$$

$$= 1 - \frac{(0.022)(4)}{2}$$

$$= 1 - 0.044$$

$$\text{Adjusted } R^2 = 0.956$$

Interpretation

After considering the number of predictors, the model explains 95.6% of the variation.

3. Mean Absolute Error (MAE)

$$MAE = \frac{|Y - \hat{Y}|}{n}$$

Absolute Errors = 2,3,2,2,1

No. of Predictors = 5

$$MAE = \frac{10}{5}$$

$$MAE = 2$$

Interpretation

On average, the model predicts marks with an error of 2 marks.

4. Mean Squared Error (MSE)

$$MSE = \frac{(Y - \hat{Y})^2}{n}$$

Residual Squares = 4,9,4,4,1

Total = 22

$$MSE = \frac{22}{5}$$

$$MSE = 4.4$$

Interpretation

The average squared prediction error is 4.4.

5. Root Mean Squared Error (RMSE)

$$RMSE = \sqrt{MSE}$$
$$= \sqrt{4.4}$$

$$RMSE = 2.10$$

Interpretation

The model's predictions differ from the actual marks by approximately 2.10 marks on average.

6. Residual Analysis

Residual Table:

Student	A	B	C	D	E
Residual	2	-3	2	-2	1

Interpretation

- Residuals are close to zero.
- Both positive and negative residuals are present.
- There is no obvious pattern in the residuals.
- Therefore, the regression model appears to fit the data well.

Interpretation and Comparison Of Regression Models

After building a regression model, it is important to interpret its performance and compare it with other models. This helps us determine which model makes more accurate predictions and best fits the data. Model interpretation involves understanding the meaning of evaluation metrics such as R^2 , Adjusted R^2 , MAE, MSE, RMSE, and Residual Analysis. Model comparison involves evaluating multiple regression models using these metrics and selecting the best-performing one.

Interpretation of Regression Models

Interpretation of a regression model is the process of analyzing evaluation metrics to determine the accuracy, reliability, and goodness of fit of the model.

Interpretation Guide

- ✓ For R^2 and Adjusted R^2 → Higher is better
- ✓ For MAE, MSE, RMSE → Lower is better
- ✓ Residuals should be random
- ✓ Model with higher R^2 /Adjusted R^2 and lower errors is preferred

Comparison of Regression Models

Comparison of regression models is the process of evaluating multiple regression models using performance metrics such as R^2 , Adjusted R^2 , MAE, MSE, RMSE, and residual analysis to select the best-performing model.

Problem: Predict the Exam Score of students based on Study Hours.

Dataset: 100 students | Target (Y): Exam Score | Predictor (X): Study Hours

Evaluation Metric	What it Measures	Model 1 (Simple Linear Regression)	Model 2 (Multiple Linear Regression)	Better Model (Why?)
R^2 (Coefficient of Determination)	Proportion of variation explained by the model (Higher is better)	0.86	0.92	Model 2 (Higher R^2)
Adjusted R^2	Explains variation after considering number of predictors (Higher is better)	0.85	0.91	Model 2 (Higher Adjusted R^2)
MAE (Mean Absolute Error)	Average absolute prediction error (Lower is better)	6.20	4.10	Model 2 (Lower MAE)
MSE (Mean Squared Error)	Average squared prediction error (Lower is better)	62.30	28.40	Model 2 (Lower MSE)
RMSE (Root Mean Squared Error)	Average prediction error in same unit as target (Lower is better)	7.89	5.33	Model 2 (Lower RMSE)
Residual Analysis	Checks if residuals are randomly distributed around zero	Some pattern seen	Randomly scattered around zero	Model 2 (Best Residuals)

INTERPRETATION

- Model 2 has higher R^2 (**0.92**) and Adjusted R^2 (**0.91**), so it explains **more** variation in exam scores.
- Model 2 has **lower MAE, MSE, and RMSE**, so its predictions are more accurate.
- Residuals of **Model 2** are randomly scattered around zero, indicating a better fit.

CONCLUSION



Model 2 (Multiple Linear Regression) is the better model for predicting exam scores because it has higher explanatory power, lower errors, and better residual behavior.

Regularization Techniques

Regularization is a machine learning technique that adds a penalty term to the regression cost function to reduce model complexity, prevent overfitting, and improve the model's ability to generalize to unseen data.

Why is Regularization Needed?

Consider a model that predicts house prices using many features:

House Size, Number of Bedrooms, Age of House, Distance from City, Parking Area, Garden Size, Wall Color, House Number

Some features like Wall Color or House Number have almost no effect on house price.

Without regularization:

- The model may assign large coefficients even to unimportant features.
- It memorizes the training data.
- Prediction accuracy on new data decreases.

With regularization:

- Important features keep reasonable coefficients.
- Less important features are reduced or removed.
- The model becomes more reliable and generalizes better.

Types of Regularization

There are two commonly used regularization techniques:

1. Ridge Regression (L2 Regularization)
2. Lasso Regression (L1 Regularization)

1. Ridge Regression (L2 Regularization)

Ridge Regression is a regularization technique that adds an L2 penalty (sum of squared coefficients) to the cost function of linear regression. It shrinks the regression coefficients toward zero without making them exactly zero, thereby reducing overfitting and improving model generalization.

Cost Function

Linear Regression

$$RSS = \sum \left(Y - \hat{Y} \right)^2$$

Ridge Regression

$$Cost = RSS + \lambda \sum_{j=1}^p \beta_j^2$$

Components

- RSS = Residual Sum of Squares
- β = Regression coefficients
- λ (Lambda) = Regularization parameter
- Ridge (L2) $\rightarrow$ Square the coefficients (β^2).

Lasso Regression (L1 Regularization)

Lasso Regression is a regularization technique that adds an L1 penalty (sum of the absolute values of the regression coefficients) to the cost function of linear regression. It shrinks the coefficients toward zero, and some coefficients become exactly zero, thereby performing feature selection and reducing overfitting.

Cost Function

Linear Regression

$$RSS = \sum \left(Y - \hat{Y} \right)^2$$

Ridge Regression

$$Cost = RSS + \lambda \sum_{j=1}^p |\beta_j|$$

Components

- RSS = Residual Sum of Squares
- β = Regression coefficients
- λ (Lambda) = Regularization parameter
- Lasso (L1) → Take the Absolute value of the coefficients ($|\beta|$).

Practical Applications of Regression

Regression is a statistical and machine learning technique used to model the relationship between dependent and independent variables for predicting continuous numerical outcomes. It is widely used in various fields to make predictions, identify relationships between variables, and support decision-making.

Advantages of Using Regression

- Predicts continuous numerical values.
- Helps identify relationships between variables.
- Supports better decision-making.
- Useful for forecasting future trends.
- Simple to understand and interpret.

In Short,

Regression is widely used in real estate, healthcare, finance, banking, marketing, education, agriculture, weather forecasting, manufacturing, transportation, insurance, sports, energy, and e-commerce to predict numerical values such as prices, sales, salaries, costs, demand, crop yield, and performance, making it one of the most important techniques in data science and machine learning.

Field		Application of Regression
	Real Estate	Predict house prices based on size, location, age, and number of rooms.
	Healthcare	Predict a patient's medical expenses, blood pressure, or recovery time.
	Finance	Forecast stock prices, company profits, or loan repayment amounts.
	Banking	Estimate loan amounts, customer income, or credit scores.
	Retail & Sales	Predict future sales based on advertising, season, and customer demand.
	Marketing	Estimate the impact of advertising expenditure on product sales.
	Education	Predict students' exam scores using study hours, attendance, and assignments.
	Agriculture	Predict crop yield based on rainfall, fertilizer use, and temperature.
	Weather Forecasting	Predict temperature, rainfall, and humidity levels.
	Manufacturing	Estimate production costs, machine efficiency, and product demand.
	Transportation	Predict travel time, fuel consumption, or traffic volume.
	Insurance	Estimate insurance premiums and claim amounts.
	Sports Analytics	Predict player performance, scores, or team rankings.

Summary

Predictive Modelling and Machine Learning are essential techniques that enable computers to learn from data and make accurate predictions. This presentation covered the complete machine learning process, including data preprocessing, model building, training, testing, evaluation, deployment, and maintenance. It explained Simple and Multiple Linear Regression, their assumptions, regression coefficients, and multicollinearity. Important evaluation metrics such as R^2 , Adjusted R^2 , MAE, MSE, RMSE, and Residual Analysis were discussed to assess model performance. The presentation also introduced Ridge and Lasso Regression to reduce overfitting and improve model generalization. These concepts provide a strong foundation for developing reliable predictive models for real-world applications.



Thank You!

Questions and Feedback are Welcome.

